1. File and Directory Management

- `ls` – List directory contents
- `cd` – Change directory
- `pwd` – Print working directory
- `cp` – Copy files and directories
- `mv` – Move or rename files and directories
- `rm` – Remove files or directories
- `mkdir` – Make directories
- `rmdir` – Remove empty directories
- `touch` – Change file timestamps or create empty files
- `find` – Search for files in a directory hierarchy
- `locate` – Find files by name
- `tree` – Display directories in a tree-like format
- `chmod` – Change file permissions
- `chown` – Change file owner and group
- `chgrp` – Change group ownership
- `stat` – Display file or file system status

2. File Viewing and Editing

- `cat` – Concatenate and display file content
- `tac` – Concatenate and display file content in reverse
- `more` – View file content interactively (page by page)
- `less` – View file content interactively (scrollable)
- `head` – Output the first part of a file
- `tail` – Output the last part of a file
- `nano` – Text editor (terminal-based)
- `vim / vi` – Advanced text editors
- `emacs` – Text editor

- `grep` – Search text using patterns
- `sed` – Stream editor for filtering and transforming text
- `awk` – Pattern scanning and processing language
- `cut` – Remove sections from each line of files
- `sort` – Sort lines of text files
  `uniq` – Report or omit repeated lines

3. Process Management

- `ps` – Report a snapshot of current processes
- `top` – Display Linux tasks
- `htop` – Interactive process viewer (advanced top)
- `kill` – Send a signal to a process, typically to terminate
- `killall` – Terminate processes by name
- `bg` – Resume a suspended job in the background
- `fg` – Bring a job to the foreground
- `jobs` – List active jobs
- `nice` – Run a program with modified scheduling priority
- `renice` – Alter priority of running processes
- `uptime` – Show how long the system has been running
- `time` – Measure program running time

4. Disk Management

- `df` – Report file system disk space usage
- `du` – Estimate file space usage
- `fdisk` – Partition table manipulator for Linux
- `lsblk` – List information about block devices
- `mount` – Mount a file system
- `umount` – Unmount a file system

- `parted` – A partition manipulation program
- `mkfs` – Create a file system
- `fsck` – File system consistency check and repair
- `blkid` – Locate/print block device attributes

5. Networking

- `ifconfig` – Configure network interfaces
- `ip` – Show/manipulate routing, devices, and tunnels
- `ping` – Send ICMP Echo requests to network hosts `netstat` – Network statistics
- `ss` – Socket statistics (faster than netstat)
- `traceroute` – Trace the route packets take to a network host
- `nslookup` – Query Internet name servers interactively
- `dig` – DNS lookup utility
- `wget` – Non-interactive network downloader
- `curl` – Transfer data with URLs
- `scp` – Secure copy files between hosts
- `ssh` – Secure shell for remote login
- `ftp` – File Transfer Protocol client

6. User and Group Management

- `useradd` – Add a user to the system
- `usermod` – Modify a user account
- `userdel` – Delete a user account
- `groupadd` – Add a group to the system
- `groupdel` – Delete a group
- `passwd` – Change user password
- `chage` – Change user password expiry information

- `whoami` – Print the current logged-in user
- `who` – Show who is logged in
- `w` – Show who is logged in and what they're doing
- `id` – Display user and group information
- `groups` – Show user's groups

7. System Information and Monitoring

- `uname` – Print system information
- `hostname` – Show or set the system's hostname
- `uptime` – How long the system has been running
- `dmesg` – Boot and system messages
- `free` – Display memory usage
- `top` – Display Linux tasks
- `vmstat` – Report virtual memory statistics
  `lscpu` – Display information about the CPU architecture
- `lsusb` – List USB devices
- `lspci` – List PCI devices
- `lshw` – List hardware configuration

8. Archiving and Compression

- `tar` – Archive files
  - tar -czf archive.tar.gz /path/to/directory – Compress files using gzip
  - tar -xzf archive.tar.gz – Extract gzipped tarball
  - tar -cf archive.tar /path/to/directory – Create a tarball
  - tar -xf archive.tar – Extract tarball
- `zip` – Package and compress files into a ZIP archive
- `unzip` – Extract files from a ZIP archive
- `gzip` – Compress files using the gzip algorithm

- `gunzip` – Decompress files compressed with gzip
- `bzip2` – Compress files using the bzip2 algorithm
- `bunzip2` – Decompress files compressed with bzip2
- `xz` – Compress files using the xz algorithm
- `unxz` – Decompress files compressed with xz

9. Package Management (Depends on Distribution)

Debian-based (e.g., Ubuntu)

- `apt-get` – APT package handling utility ◦ apt-get install <package> – Install a package ◦ apt-get update – Update package list ◦ apt-get upgrade – Upgrade installed packages ◦ apt-get remove <package> – Remove a package
- `apt-cache` – Query APT cache ◦ apt-cache search <package> – Search for a package ◦ apt-cache show <package> – Show package details

Red Hat-based (e.g., CentOS, Fedora)

`yum` – Package manager for RPM-based systems ◦ yum install <package> – Install a package ◦ yum update – Update installed packages ◦ yum remove <package> – Remove a package

- `dnf` – Next-generation package manager (Fedora, CentOS 8+) ◦ dnf install <package> – Install a package ◦ dnf update – Update installed packages ◦ dnf remove <package> – Remove a package

General Commands

- `rpm` – RPM package manager ○ rpm -i <package.rpm> – Install an RPM package ○ rpm -e <package> – Remove an RPM package
- `dpkg` – Debian package manager ○ dpkg -i <package.deb> – Install a Debian package ○ dpkg -r <package> – Remove a Debian package

10. System Services and Daemon Management

- `systemctl` – Control the systemd system and service manager ○ systemctl start <service> – Start a service ○ systemctl stop <service> – Stop a service ○ systemctl restart <service> – Restart a service ○ systemctl enable <service> – Enable a service to start on boot ○ systemctl disable <service> – Disable a service from starting on boot
  ○ systemctl status <service> – Check service status
- `service` – Older service management command (used in nonsystemd systems) ○ service <service> start – Start a service ○ service <service> stop – Stop a service ○ service <service> restart – Restart a service ○ service <service> status – Check service status

11. Scheduling Tasks

- `cron` – Daemon for running scheduled commands ◦ crontab -e – Edit cron jobs for the current user ◦ crontab -l – List the current user's cron jobs ◦ crontab -r – Remove the current user's cron jobs
- `at` – Run commands at a specified time ◦ at 09:00 – Schedule a command to run at 09:00 AM
- `batch` – Run commands when the system load is low
- `sleep` – Delay for a specified time ◦ sleep 5s – Sleep for 5 seconds

12. File Permissions and Security

- `chmod` – Change file permissions
- `chown` – Change file owner and group
- `chgrp` – Change the group ownership of a file
- `umask` – Set default permissions for new files
- `setfacl` – Set file access control lists (ACL)
- `getfacl` – Get file access control lists (ACL)
- `sudo` – Execute a command as another user (usually root)
- `visudo` – Edit the sudoers file safely
- `passwd` – Change a user's password
- `sudoers` – Manage sudo access for users
- `gpasswd` – Administer group password
- `ss` – Display socket statistics (for secure network connections)

13. System Backup and Restore

- **rsync** – Remote file and directory synchronization ○ rsync -avz source/ destination/ – Synchronize files ○ rsync -avz -e ssh source/ user@remote:/destination/ – Sync over SSH
- **cpio** – Copy files to and from archives
- **dd** – Low-level copying and backup of entire filesystems
  - dd if=/dev/sda of=/path/to/backup.img – Backup a disk/partition
  - dd if=/path/to/backup.img of=/dev/sda – Restore a disk/partition

14. System Diagnostics and Troubleshooting

- **dmesg** – Print the kernel ring buffer messages (system boot and hardware-related messages)
- **journalctl** – Query and view logs from systemd's journal
- **strace** – Trace system calls and signals ○ strace <command> – Trace a command's system calls
- **lsof** – List open files (useful for debugging) ○ lsof <file> – Show processes using a specific file
- **vmstat** – Report virtual memory statistics
- **iostat** – Report CPU and I/O statistics
- **mpstat** – Report CPU usage statistics
- **pidstat** – Report statistics by process
- **free** – Display memory usage
- **uptime** – How long the system has been running
- **watch** – Execute a program periodically, showing output ○ watch -n 1 free – Watch memory usage every second
- **lshw** – List hardware configuration
- **htop** – Interactive process viewer (better than top)

- `netstat` – Network statistics (deprecated in favor of ss)
- `ss` – Show socket statistics (more efficient than netstat)

15. Networking & Remote Management

- `ifconfig` – Configure network interfaces (older command, replaced by ip)
- `ip` – A more modern alternative for managing network interfaces and routing
  - ip addr – Show IP addresses
    - ip link – Show or manipulate network interfaces
    - ip route – Show or manipulate routing tables
- `ss` – Display socket statistics (useful for diagnosing network issues)
- `nmap` – Network exploration tool (can be used for security auditing)
- `telnet` – User interface to the TELNET protocol (less common nowadays)
- `nc` (Netcat) – Network utility for reading and writing from network connections
  - nc -l -p 1234 – Listen on port 1234
    - nc <host> <port> – Connect to a host and port
- `iptables` – Administration tool for IPv4 packet filtering and NAT (Network Address Translation)
- `firewalld` – Frontend for managing firewall rules (used in some distros like Fedora and CentOS)
- `ufw` – Uncomplicated firewall (front-end for iptables)
  - ufw enable – Enable firewall
    - ufw allow <port> – Allow traffic on a specific port
- `tcpdump` – Command-line packet analyzer
- `curl` – Transfer data from or to a server using various protocols (HTTP, FTP, etc.)
- `wget` – Download files from the web via HTTP, HTTPS, FTP

- **scp** – Secure copy over SSH (used to copy files between systems)
  - scp file.txt user@remote:/path/to/destination/ – Copy file to remote server
- **rsync** – Remote file and directory synchronization (often used for backups)
  - rsync -avz /local/path/ remote:/remote/path/ – Sync directories

16. Text Processing Utilities

- **grep** – Search for patterns within files
  - grep 'pattern' file.txt – Search for a pattern in a file
  - grep -r 'pattern' /dir/ – Recursively search for a pattern
- **sed** – Stream editor for filtering and transforming text
  - sed 's/old/new/g' file.txt – Replace old with new globally
- **awk** – A powerful text processing language
  - awk '{print $1}' file.txt – Print the first column of each line in a file
- **cut** – Remove sections from each line of a file
  - cut -d ':' -f 1 /etc/passwd – Print the first field of each line, delimited by ":"
- **sort** – Sort lines of text files
  - sort file.txt – Sort file content in ascending order
- **uniq** – Report or omit repeated lines in a file
  - sort file.txt | uniq – Sort and remove duplicate lines
- **tee** – Read from standard input and write to standard output and files
  - echo "text" | tee file.txt – Write to file and show output on screen
- **tr** – Translate or delete characters
  - echo "hello" | tr 'a-z' 'A-Z' – Convert lowercase to uppercase
- **paste** – Merge lines of files
  - paste file1.txt file2.txt – Combine lines of file1 and file2 side by side

- **wc** – Word, line, character, and byte count ○ wc -l file.txt – Count lines in a file ○ wc -w file.txt – Count words in a file

17. System Shutdown and Reboot

- **shutdown** – Shut down the system ○ shutdown -h now – Immediately shut down ○ shutdown -r now – Reboot the system ○ shutdown -h +10 – Shut down after 10 minutes
- **reboot** – Reboot the system
- **halt** – Halt the system immediately (equivalent to turning off power)
- **poweroff** – Power off the system

- `init` – Change the runlevel (old-style system manager) ◦ init 0 – Shutdown ◦ init 6 – Reboot

18. File System Mounting and Management

- `mount` – Mount a file system ◦ mount /dev/sda1 /mnt – Mount partition to a directory
- `umount` – Unmount a file system ◦ umount /mnt – Unmount the file system mounted at /mnt
- `fstab` – File system table (configuration file for mounting file systems) ◦ /etc/fstab – View and configure persistent mount points
- `blkid` – Display block device attributes
- `fsck` – Check and repair a file system ◦ fsck /dev/sda1 – Check and repair /dev/sda1

19. Filesystem Permissions and Security

- `chmod` – Change file permissions ◦ chmod 755 file.txt – Give read, write, and execute permissions to owner, and read-execute permissions to others
- `chown` – Change file owner and group ◦ chown user:group file.txt – Change owner and group of a file
- `chgrp` – Change group ownership of a file ◦ chgrp group file.txt – Change the group of a file
- `umask` – Set default permissions for new files ◦ umask 022 – Set default permissions for newly created files to 755
- `setfacl` – Set access control lists (ACL) for file permissions
- `getfacl` – Get access control lists (ACL) for file permissions

20. Containerization and Orchestration

Docker

- `docker` – Docker command-line interface (CLI) for managing containers
  - `docker run <image>` – Run a container from an image
  - `docker ps` – List running containers
  - `docker ps -a` – List all containers, including stopped ones
  - `docker build -t <image_name> .` – Build an image from a Dockerfile
  - `docker exec -it <container_id> bash` – Start an interactive bash shell inside a running container
  - `docker stop <container_id>` – Stop a container
  - `docker rm <container_id>` – Remove a container
  - `docker logs <container_id>` – View logs of a container
  - `docker images` – List available images
  - `docker rmi <image_name>` – Remove an image
  - `docker network ls` – List Docker networks
  - `docker-compose` – Manage multi-container Docker applications
    - `docker-compose up` – Start up a multi-container environment
    - `docker-compose down` – Stop and remove containers created by docker-compose
    - `docker-compose logs` – View logs from containers managed by docker-compose

Kubernetes (k8s)

- `kubectl` – Command-line tool for interacting with Kubernetes clusters

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- `kubectl get pods` – List pods in the current namespace
- `kubectl get nodes` – List nodes in the cluster
- `kubectl get services` – List services in the cluster
- `kubectl apply -f <file>.yaml` – Apply configuration from a file (e.g., a deployment or pod configuration)
- `kubectl create -f <file>.yaml` – Create a resource from a file
- `kubectl delete -f <file>.yaml` – Delete a resource defined in a file
- `kubectl exec -it <pod_name> -- bash` – Execute a command inside a pod (e.g., open a shell)
- `kubectl logs <pod_name>` – View the logs of a pod
- `kubectl describe pod <pod_name>` – Get detailed information about a pod
- `kubectl scale deployment <deployment_name> -replicas=<number>` – Scale a deployment to the desired number of replicas
- `kubectl rollout restart deployment <deployment_name>` – Restart a deployment
- `kubectl port-forward pod <pod_name> <local_port>:<remote_port>` – Forward a port from a pod to localhost

Helm

- `helm` – Kubernetes package manager for deploying applications
  - `helm install <release_name> <chart_name>` – Install a Helm chart
  - `helm upgrade <release_name> <chart_name>` –

Upgrade a Helm release ◦ `helm list` – List all Helm releases
◦ `helm delete <release_name>` – Delete a Helm release ◦ `helm search <chart_name>` – Search for a Helm chart

21. Automation and Configuration Management

Ansible

- `ansible` – Automation tool for configuration management

- `ansible all -m ping` – Ping all hosts defined in the inventory
- `ansible-playbook playbook.yml` – Run an Ansible playbook
- `ansible -m command -a 'command' <host>` – Run a single command on a target host
- `ansible-playbook --check playbook.yml` – Dry-run a playbook to see what would change
- `ansible-playbook --limit <host> playbook.yml` – Run a playbook on a specific host or group
- `ansible-playbook --extra-vars "key=value"` – Pass extra variables to a playbook

Terraform

- `terraform` – Infrastructure as code tool for provisioning and managing cloud resources
  - `terraform init` – Initialize a working directory for Terraform configuration
  - `terraform plan` – Show an execution plan (preview of what changes will be made)
  - `terraform apply` – Apply the changes described in a Terraform configuration
  - `terraform destroy` – Destroy infrastructure created by Terraform
  - `terraform validate` – Validate the configuration files
  - `terraform show` – Show the current state of the infrastructure

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Puppet

- `puppet` – Configuration management tool
  - `puppet apply <manifest.pp>` – Apply a Puppet manifest locally
  - `puppet agent --test` – Test the Puppet agent (can be used to run a one-off run)
  - `puppet resource` – Show the current state of resources (files, services, etc.)

22. CI/CD Tools and Commands

Jenkins

- `jenkins` – Continuous integration tool
  - `java -jar jenkins.war` – Start Jenkins from a WAR file
  - Access Jenkins through http://localhost:8080 by default

GitLab CI

- `.gitlab-ci.yml` – Configuration file for GitLab CI/CD pipelines (typically resides in your repository)
  - `gitlab-runner register` – Register a new runner with GitLab
  - `gitlab-runner run` – Run the GitLab Runner to process jobs

GitHub Actions

- GitHub Actions uses YAML configuration files (typically located in .github/workflows/)
  - `actions/checkout@v2` – Checkout the repository code in your CI pipeline

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- `actions/setup-node@v2` – Setup Node.js for use in a pipeline
- `docker/setup-buildx-action@v1` – Set up Docker Buildx for building multi-platform images

23. Cloud Services

AWS CLI (Amazon Web Services)

- `aws` – Command-line tool for managing AWS services
  - `aws configure` – Configure AWS CLI with your credentials
  - `aws s3 cp file.txt s3://bucket-name/` – Copy a file to an S3 bucket
  - `aws ec2 describe-instances` – Describe EC2 instances
  - `aws ec2 start-instances --instance-ids <id>` – Start an EC2 instance
  - `aws ec2 stop-instances --instance-ids <id>` – Stop an EC2 instance
  - `aws s3 sync` – Sync directories with an S3 bucket

Azure CLI (Microsoft Azure)

- `az` – Command-line tool for managing Azure services
  - `az login` – Log in to your Azure account
  - `az vm list` – List all virtual machines
  - `az vm start --name <vm_name> --resource-group <resource_group>` – Start an Azure VM
  - `az storage blob upload` – Upload files to an Azure blob storage
  - `az group create` – Create a new resource group in Azure

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Google Cloud SDK (gcloud)

- `gcloud` – Command-line tool for Google Cloud Platform
  - `gcloud auth login` – Log in to Google Cloud
  - `gcloud compute instances list` – List compute instances
  - `gcloud compute instances stop <instance_name>` – Stop a Google Cloud VM instance
  - `gcloud app browse` – Open the current Google App Engine application in a browser

24. Logging and Monitoring

Prometheus

- `prometheus` – Open-source system monitoring and alerting toolkit
  - `prometheus` – Start Prometheus server (usually runs as a service in the background)
    - `prometheus --config.file=<config_file>` – Start Prometheus with a specific config file

Grafana

- `grafana-cli` – Command-line interface for managing Grafana plugins
  - `grafana-cli plugins install <plugin-name>` – Install a plugin in Grafana

ELK Stack (Elasticsearch, Logstash, Kibana)

- `elasticsearch` – Search engine for logging and data analytics
  - `curl -XGET 'localhost:9200/_cluster/health?pretty'` – Get cluster health status
- `logstash` – Server-side data processing pipeline
  - `logstash -f <config_file>` – Run Logstash with the specified configuration file
- `kibana` – Web interface for visualizing Elasticsearch data
  - Kibana is generally accessed through a web browser (http://localhost:5601)